

Mohammad Amin Shirazian

Research Statement

Department of Economics, Toronto Metropolitan University

✉ amin.shirazian.work@gmail.com · 🌐 Website · 🔗 LinkedIn

I am a macroeconomist interested in how health shocks and incomplete insurance markets shape household saving, wealth inequality, and welfare. Methodologically, I combine economic theory, household survey data, and quantitative macroeconomic models. My research focus is on why precautionary saving remains active at the top of the wealth distribution and whether households across the wealth distribution save optimally in response to health shocks.

Current Work - Job Market Paper

My current work addresses three connected questions. First, what are the sufficient conditions under which the precautionary motive to self-insure against health shocks remains active at arbitrarily high wealth? Second, what is the health production function, what parameters govern it, and how do they depend on the existing health system and available medical technologies? Third, how much of the observed saving at the top of the wealth distribution can the precautionary motive to self-insure against health shocks explain, given the existing health system and available medical technologies? I discuss each question in turn.

To address the first question, I start with an income-fluctuation model with an unspecified low state and establish a lemma that gives the criterion for positive saving in terms of the marginal continuation value across states. I then incorporate the Hall and Jones (2007) framework into the low state and derive the sufficient condition under which the lemma applies. My main theorem establishes that, under the maintained concavity assumption, saving is positive beyond a wealth threshold when the marginal utility of consumption declines faster than the marginal return on survival. I then show that, together with random separation through mortality, positive saving at high wealth generates a stationary wealth distribution with a Pareto upper tail. Future work will study other examples in which the same precautionary mechanism can remain active at arbitrarily high wealth.

To address the second question, I estimate a health production function in which medical spending improves survival. The estimates imply that, for pre-retirement households that have experienced a health shock, every \$100 of private medical spending produces approximately \$1.36 of effective additional care beyond the care already provided through other sources. They also imply that a 1 percent increase in total effective health input raises health output by approximately 0.39 percent, so the return to medical spending remains positive but diminishes as spend-

ing increases. Future work will study longevity-enhancing health products directly, including their effectiveness and prices, to develop a richer health production function.

To address the third question, I use the estimated health production parameters to calibrate the model to pre-retirement couple households in the United States. For households in the top wealth decile, the model predicts average saving of approximately \$122,000, compared with approximately \$1.09 million in the data. The precautionary motive to self-insure against health shocks therefore accounts for approximately 11 percent of observed top-decile saving under the baseline calibration and closes nearly \$2 million of the gap relative to the case without health shocks. Future work will incorporate a bequest motive and family dynamics to determine what share of observed saving can be explained by the extended model.

Research Agenda & Future Projects

My research agenda asks whether we correctly understand how health shocks and incomplete insurance markets shape household saving and whether the resulting saving decisions are socially optimal. My job-market paper shows that health shocks can sustain precautionary saving among families at the top of the wealth distribution, where standard models predict that this motive should have faded. This finding raises a broader question: do wealthy households save too much, or do poorer households save too little? The answer depends partly on how effectively private medical spending supplements existing care and on the range and diminishing returns of the medical technologies available to households. Building on my current work, I plan to estimate how these features vary across households and healthcare systems, study which private health products improve longevity and how closely they substitute for publicly provided care, and evaluate whether government intervention can improve the allocation of healthcare and insurance. I organize this agenda around two future projects.

The first project will study the health production function by examining longevity-enhancing health products and medical technologies directly. I will identify which products improve survival, how effectively private treatment supplements the care already available, and how rapidly the returns to additional medical spending diminish. I will then examine the wealth threshold above which households begin to demand these products and the household characteristics, beyond wealth, that shape the demand for longevity. A central empirical challenge is that wealth may increase the desire to extend life, while a stronger desire for longevity may itself induce greater wealth accumulation.

The second project will extend my current framework to a general-equilibrium model with bequest motives and family dynamics. The model will capture how households share health risks within families, provide informal care, transfer wealth across generations, and save for both future medical needs and bequests. I will use the extended framework to determine how much of observed saving across the wealth distribution can be explained by these additional motives and to evaluate whether household decisions under the existing healthcare and insurance system lead to an efficient allocation of resources. This comparison will identify when incomplete insurance markets create a rationale for government intervention and will allow me to study the welfare effects and design of alternative health insurance policies.

References

Robert E. Hall and Charles I. Jones. The value of life and the rise in health spending. *The Quarterly Journal of Economics*, 122(1):39–72, 2007. doi: 10.1162/qjec.122.1.39.